

ORLINGA, Russian Federation

WMO: 303280

Lat: 56.050N

Lon: 105.833E

Elev: 1109

StdP: 14.12

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.5	-41.6	-49.7	0.3	-45.3	-45.3	0.4	-40.9	13.0	0.6	11.7	2.1	0.0	250	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	83.9	64.7	80.2	63.4	76.5	61.8	67.6	79.0	66.0	76.4	64.2	73.5	3.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	93.1	71.5	62.2	87.1	70.1	60.5	82.0	68.4	32.8	78.8	31.5	76.3	30.1	73.7	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.3	10.8	8.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-49.3	69.6	3.9	2.6	-52.1	71.5	-54.4	73.1	-56.6	74.5	-59.5	76.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	-15.1	-5.6	10.5	31.2	44.7	59.5	64.4	57.9	44.4	30.0	6.0	-10.2		
	DBStd	29.50	15.15	14.03	12.33	8.53	6.63	6.71	5.07	5.79	7.08	9.05	16.25	16.37		
	HDD50	9573	2019	1557	1224	563	192	8	0	9	196	621	1319	1865		
	HDD65	14082	2484	1977	1689	1013	629	187	74	226	618	1086	1769	2330		
	CDD50	1050	0	0	0	0	28	293	445	255	29	0	0	0		
	CDD65	83	0	0	0	0	0	22	54	7	0	0	0	0		
	CDH74	1292	0	0	0	0	24	466	668	122	12	0	0	0		
	CDH80	324	0	0	0	0	4	112	182	24	2	0	0	0		
	WSAvg	2.3	1.6	1.5	1.9	3.2	3.5	2.5	2.2	2.0	2.4	2.6	2.5	1.9		
	PrecAvg	18.1	0.8	0.6	0.6	0.7	1.3	2.2	3.1	3.2	1.8	1.4	1.1	1.0		
(15) Precipitation	PrecMax	23.6	1.6	1.2	1.5	2.1	2.4	4.1	7.6	5.5	3.2	2.7	1.8	2.0		
	PrecMin	13.0	0.3	0.2	0.1	0.2	0.2	0.4	0.7	1.8	0.1	0.2	0.6	0.4		
	PrecStd	2.8	0.4	0.3	0.4	0.4	0.6	0.9	1.5	1.1	0.7	0.6	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	30.9	46.6	59.7	78.2	87.8	89.6	82.6	73.7	55.9	38.4	28.1		
		MCWB	20.6	27.1	35.9	44.5	55.6	66.1	68.2	64.9	59.3	46.0	34.3	26.3		
	2%	DB	16.3	24.4	39.9	54.1	71.6	83.3	85.2	76.9	64.8	48.8	34.2	21.1		
		MCWB	15.1	21.8	31.7	41.9	52.9	64.0	65.7	62.9	53.7	42.1	31.7	19.3		
	5%	DB	11.1	18.9	34.5	49.7	65.9	79.7	81.3	73.0	60.1	44.6	31.0	17.5		
		MCWB	10.0	17.0	28.7	39.1	49.7	62.0	64.7	61.5	51.2	40.3	28.7	16.3		
	10%	DB	5.9	14.7	30.6	44.9	60.4	75.9	77.6	69.5	56.0	40.7	27.3	13.3		
		MCWB	4.9	13.2	26.1	36.2	47.6	60.2	63.7	60.1	49.3	37.0	25.5	12.3		
	0.4%	WB	20.7	27.4	36.0	45.9	58.2	68.2	70.6	67.5	60.8	46.7	34.9	26.2		
		MCDB	21.8	30.8	44.9	57.3	73.9	81.4	83.0	79.7	73.7	52.4	36.8	28.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	15.3	21.8	32.3	42.6	53.9	66.2	68.5	65.1	56.1	43.4	31.8	19.5		
		MCDB	16.5	24.0	38.7	52.6	68.6	79.4	79.4	73.9	62.5	47.9	34.1	20.9		
	5%	WB	10.0	17.2	29.1	39.5	51.1	64.2	67.0	63.2	52.8	40.6	28.7	16.4		
		MCDB	11.0	18.9	34.2	48.0	63.9	75.9	77.4	70.7	57.9	44.4	30.8	17.5		
	10%	WB	5.0	13.3	26.2	37.0	48.5	61.9	65.5	61.0	50.2	37.3	25.6	12.4		
		MCDB	5.9	14.7	30.9	43.6	59.1	72.1	75.0	67.3	55.2	40.1	27.4	13.3		
	5% DB	MDBR	16.5	23.2	30.9	22.4	24.2	25.7	23.0	19.9	18.4	14.2	14.9	15.3		
		MCDBR	16.3	21.6	33.2	29.1	36.8	34.4	32.0	27.2	26.7	20.1	13.8	14.3		
		MCWBR	15.7	19.3	25.4	17.1	19.2	16.0	14.6	13.6	15.1	13.2	11.5	13.4		
		MCWBR	15.7	21.2	32.0	26.5	32.2	29.7	26.1	23.7	23.9	18.8	13.1	14.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	15.0	19.2	24.8	15.9	18.3	14.4	13.5	13.2	14.1	12.7	11.3	13.4		
	taub		0.279	0.312	0.335	0.367	0.420	0.451	0.481	0.411	0.349	0.327	0.293	0.272		
		taud	2.002	1.989	2.015	2.097	2.148	2.160	2.095	2.286	2.425	2.309	2.068	1.968		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		184	227	256	267	259	252	240	252	256	227	177	151		
		Edh at Noon	21	32	40	43	44	44	46	36	27	23	20	17		
	RadAvg		263	573	1012	1407	1637	1803	1662	1373	895	477	251	174		
(45) All-Sky Solar Radiation	RadStd		24	39	53	104	131	141	166	95	83	59	29	22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page